

Introduction

Type 2 Diabetes Mellitus (T2DM) is a chronic metabolic disorder and a component of metabolic syndrome (MS), characterized by reduced insulin secretion following the development of insulin resistance (Lu et al., 2024). Nationally, T2DM represents a substantial public health crisis, affecting an estimated 38.4 million Americans, which equates to approximately 11.6% of the U.S. population (Parker et al., 2024). The burden of this condition is profound, causing extensive morbidity, disability, and mortality. T2DM is a leading cause of severe complications, including cardiovascular disease, end-stage renal disease (ESRD), nerve damage (neuropathy), retinopathy, and diabetic foot ulcers, significantly reducing life expectancy and quality of life (Lu et al., 2024). The economic impact is staggering: the total estimated economic cost of diagnosed diabetes in the U.S. in 2022 was 412.9 billion, with direct medical costs totaling 332.8 billion (Parker et al., 2024). This cost burden, along with reduced productivity, underscores the disease's immense societal strain and its critical relevance to national public health goals, such as those aimed at reducing chronic disease prevalence.

The burden of T2DM in Kings County (Brooklyn), New York, is disproportionately high, particularly within specific demographic groups, mirroring and often exceeding national health disparities. The prevalence of T2DM is highest among Black and Latino communities, where historical inequities in housing, employment, and education have concentrated poverty and limited access to resources. In Kings County, where poverty rates exceed 30% in some areas, the disease substantially reduces quality of life and increases disability through severe complications like chronic kidney disease and diabetic foot ulcers. For individuals with unfavorable social conditions, the life expectancy may be shortened by approximately three to four years at age 45

compared to those with favorable conditions (McLaurin et al., 2024). This disparity is exacerbated by the economic burden: while healthcare costs for T2DM patients are 2.3 times higher than for non-diabetic patients (Parker et al., 2024), low-income and uninsured residents in Brooklyn often face delayed care and emergency department overutilization, creating financial strain that perpetuates cycles of poverty and limits adherence to complex care regimens (Walker et al., 2014).

The risk of developing and the difficulty in managing T2DM are influenced by a convergence of modifying factors. Biological factors include family history, genetic predisposition, and co-existing conditions such as obesity, hypertension, dyslipidemia, and metabolic dysfunction associated with steatotic liver disease (MASLD) (Lu et al., 2024). These biological risks are heavily compounded by behavioral factors like a sedentary lifestyle and poor dietary patterns characterized by high sugar, saturated fat, and low fiber intake (Chen et al., 2025; Papamichou et al., 2019; Schwingshackl et al., 2017). However, these individual choices are largely shaped by social and environmental factors, which are the fundamental drivers of risk. Social determinants of health (SDOH)—such as chronic stress, low-income environments, limited access to healthy foods (Walker et al., 2014)—and poor healthcare access are critical modifiers (Levy et al., 2022). Furthermore, systemic inequities, including structural racism and provider bias, create conditions that disproportionately expose minority groups to risk while simultaneously limiting their access to high-quality care, thereby accelerating the onset and severity of T2DM (Anim et al., 2024; Titus & Kataoka-Yahiro, 2019).

Background: Systems Analysis of Health Determinants

Natural History of Disease

Type 2 diabetes begins because of an increase in insulin resistance. During the preclinical phase, although patients do not present noticeable symptoms, the body's cells do not effectively produce insulin. The clinical phase occurs when the pancreatic cells try to make more insulin but eventually exhausts itself when the body itself does not have sufficient insulin (Nishida & Watada, 2024). Type 2 diabetes is significantly influenced by genetics and lifestyle factors. Some of the ways diabetes is tested and measured is by insulin resistance levels (Cooper et al., 2025), blood glucose levels (Nishida & Watada, 2024), hemoglobin A1C test, oral glucose tolerance test (U.S. National Library of Medicine, n.d.-b), and biomarkers (Cooper et al., 2025). Diabetes is diagnosed when: fasting blood glucose level is 126 mg/dL (7.0 mmol/L) or higher; hemoglobin A1C test result is 6.5% or higher; or the oral glucose tolerance test is 200 mg/dL (11.1 mmol/L) or higher 2 hours after drinking a special glucose drink. Complications of diabetes include eye disease, kidney disease, heart disease, and stroke (U.S. National Library of Medicine, n.d.-b). The demographics at higher risk for type 2 diabetes include being overweight, middle-aged to older people, prediabetes, physically inactive, genetics, and family history (U.S. National Library of Medicine, n.d.-a).

Nutritional Factors

A systematic review and meta-analysis by Schwingshackl et al. (2017) examined the relationship between the intake of 12 major food groups and the risk of developing type 2 diabetes mellitus (T2DM). The study revealed that while red and processed meats, as well as sugar-sweetened beverages, were associated with an increased risk of T2DM, higher consumption of whole grains, fruits, and dairy correlated with a decreased risk. The authors accounted for various confounding factors such as age, sex, and BMI, yet acknowledged that other lifestyle factors might still influence the outcomes. Similarly, Pollakova et al. (2021) focused on the impact of a vegan diet on T2DM, finding that vegan dietary patterns are linked to lower diabetes prevalence and improved glycemic control. Despite considering multiple potential confounders, their review highlighted challenges in relying on self-reported dietary data and variability in vegan diet definitions, complicating comparative analyses across studies. Together, these reviews underscore the significance of dietary patterns in T2DM risk and the need for careful assessment of dietary habits and food quality in future research.

Environmental Factors

In a meta-analysis conducted by Yang et al. (2019), the long-term effects of air pollution on the incidence of type 2 diabetes mellitus (T2DM) were assessed across a total of 16 studies from countries including the USA, Canada, and several European nations. The findings revealed a significant association between increased exposure to particulate matter (PM_{2.5} and PM₁₀) and nitrogen dioxide (NO₂) with the risk of developing T2DM, particularly in American cohorts. Interestingly, the results indicated that the increased incidence was not similarly observed in European cohorts. While the analysis controlled for key confounding variables such as age, sex,

and socioeconomic status, potential unmeasured factors, including individual genetic predispositions and healthcare access, were noted as limitations. Additionally, the study's focus predominantly on urban settings may limit the generalizability of its findings to rural populations.

Social Factors

Social determinants of health (SDOH), exert a profound and measurable influence on clinical outcomes for individuals with Type 2 Diabetes Mellitus (T2DM). Systematic review evidence demonstrates that SDOH, encompassing economic stability, education, neighborhood characteristics, and social/community context, significantly affect critical health markers like glycemic control (HbA1c), LDL cholesterol, and blood pressure (Walker et al., 2014). This highlights that factors outside the traditional clinical setting—such as where a person lives, learns, works, and their economic security—are major contributors to a patient's ability to manage a complex chronic disease. While the link is clear, the measurement of outcomes like quality of life and healthcare costs in relation to these determinants remains an area needing further exploration (Walker et al., 2014).

Including the effects of adverse SDOH, racial and ethnic disparities persist in the quality of diabetes care received, even after accounting for some socioeconomic differences. Observational studies have found that racial/ethnic minority groups—including Hispanics, Blacks, and Asians—receive key quality care indicators (such as HbA1c tests, annual foot exams, and flu vaccinations) at lower rates compared to non-Hispanic Whites (Canedo et al., 2018). While controlling for insurance coverage and education level somewhat mitigates these differences, disparities in receiving HbA1c tests persisted for all three minority groups (Canedo et al., 2018).

The fact that these disparities remain, even when controlling for factors like income, suggests that systemic issues beyond simple access and poverty—including implicit bias and structural barriers—actively contribute to an inequitable standard of care for minority populations (Canedo et al., 2018).

Healthcare Access Factors

Healthcare access factors play a critical role in determining health outcomes, particularly for chronic conditions like type 2 diabetes mellitus (T2DM). Studies have shown that socioeconomic status, including income and education levels, significantly affects individuals' access to necessary healthcare services. Disparities in healthcare access can lead to lower adherence to diabetes care indicators, such as HbA1c testing, foot examinations, and vaccinations, especially among racial and ethnic minority groups. For instance, Canedo et al. (2018) found that despite controlling for various confounding variables, disparities in care persisted for certain groups, highlighting the profound impact of systemic barriers on healthcare quality. Furthermore, factors such as insurance coverage and structural racism can complicate access, resulting in poorer health outcomes and exacerbating existing health inequities. Consequently, addressing these healthcare access factors is essential for improving diabetes management and reducing disparities in T2DM outcomes.

Impacts of Bias, Inequities, and Racism

Bias, structural inequities, and racism demonstrably impact health outcomes, as evidenced by systematic reviews focusing on Type 2 Diabetes Mellitus (T2DM) in the United States.

Perceived interpersonal discrimination and racism experienced from healthcare professionals are specifically associated with negative T2DM outcomes among Black Americans (Anim et al., 2024). Provider-level factors, including communication quality and assumptions, were found to significantly mediate these negative effects, influencing patient self-management behaviors and ultimately clinical results (Anim et al., 2024). This suggests that biases held by healthcare staff and the resulting poor quality of care contribute directly to disparities in chronic disease management.

Furthermore, structural racism and broader social determinants of health are recognized as key drivers of poor T2DM outcomes. While nonmedical, multicomponent interventions that address social determinants show promise in improving clinical markers like Hemoglobin A1c (HbA1c) levels (Egede et al., 2022), the current literature addressing structural racism through actionable nonmedical interventions remains limited (Egede et al., 2022). These findings underscore that simply providing medical care is insufficient; addressing the systemic and interpersonal factors—bias, inequities, and racism—embedded within society and the healthcare system is critical for reducing disparities and improving chronic disease management, particularly for marginalized racial groups.

Statistical Methods

Background and Data Description

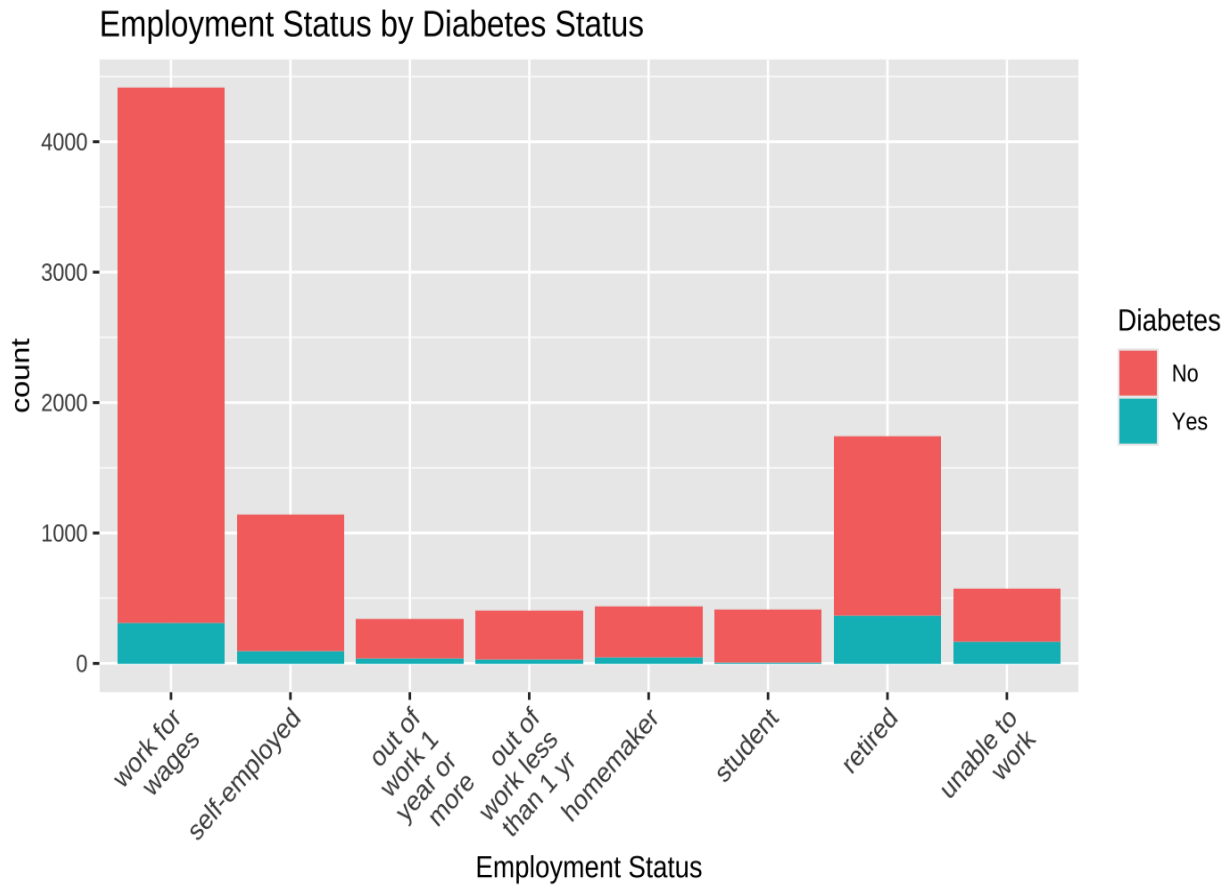
BRFSS, which stands for Behavioral Risk Factor Surveillance System, is a system of health-related telephone surveys that collect state data about adult US residents who are 18 years or older. Being one of the largest health survey systems in the world, BRFSS conducts more than 400,000 interviews yearly. Data gets collected by in-house interviews, contact by telephone call centers, or universities that administer the BRFSS surveys. It examines health-related risk behaviors, chronic health conditions, and the use of preventive services. BRFSS is important because it collects data on important health issues such as vaccine shortages and influenza illnesses. In BRFSS, diabetes is measured by a self-reported survey “Have you ever been told by a doctor that you have diabetes?” as a binary categorical answer of yes or no. Missing data was omitted from the dataset to better analyze the relationship between diabetes and various confounders.

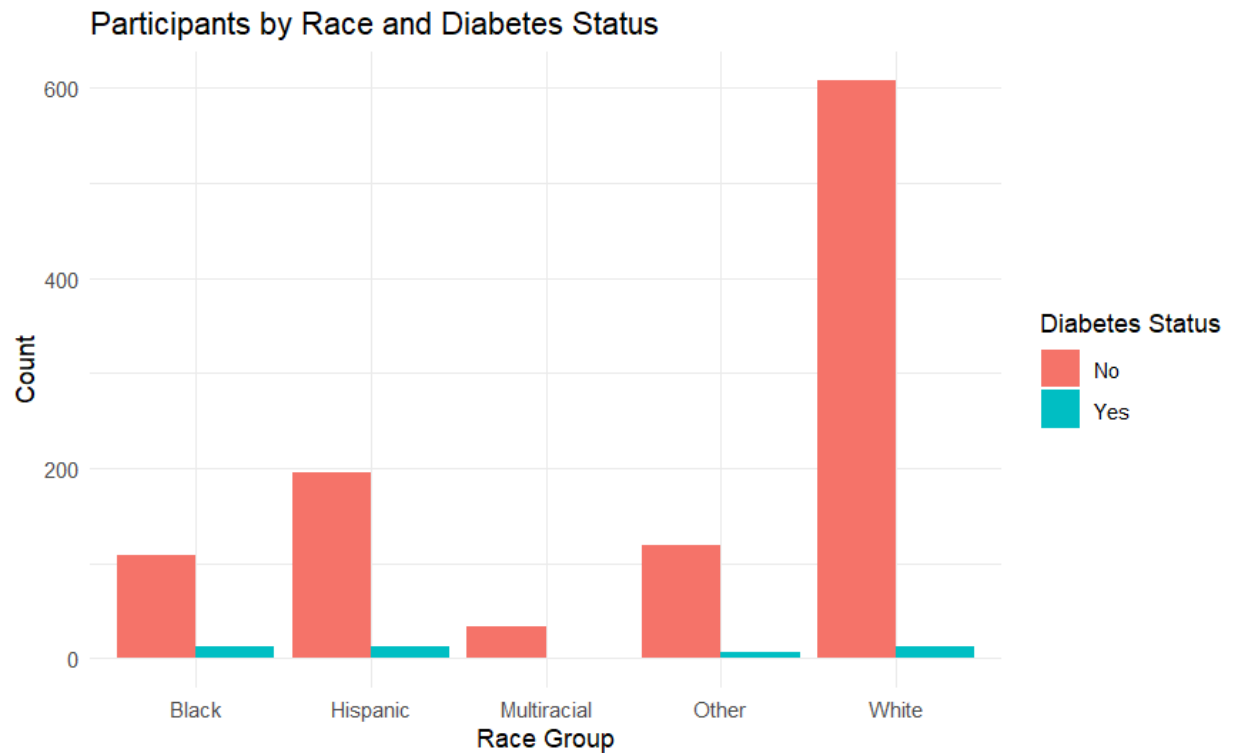
One key limitation of the BRFSS dataset is related to its sampling methodology. Since data is collected via telephone interviews, it may exclude individuals without access to phones or those who are less likely to respond to surveys, such as certain low-income populations or younger adults. This can lead to a biased sample that may not fully represent the diversity of the population. Additionally, the reliance on self-reported data introduces potential measurement bias, particularly in responses related to sensitive conditions like diabetes, where individuals may underreport or misinterpret their health status. Furthermore, the cross-sectional nature of the data limits causal inference, as it captures a snapshot in time rather than tracking changes over time or

establishing direct cause-and-effect relationships. Thus cannot establish causality because cross-sectional doesn't fit the temporality criteria of Hill's.

The main outcome variable in this analysis is Diabetes, measured by the variable DIABETE4. This is a categorical (dichotomous) variable defined by the respondent's self-report of having been told they have diabetes ("yes" or "no"). For the analysis, responses that were not clear (e.g., "Don't Know/Not Sure" or "Refused") were treated as missing data and excluded to ensure a clean definition of the outcome. The primary variables used for the disparity analysis are all categorical. These include _RACEGR3 (five-level race/ethnicity), which is nominal and coded into five groups (1- White only, 2 - Black only, 3 - Other race only, 4 - Multiracial, and 5 - Hispanic); INCOME3 (income level), an ordinal variable with 11 ascending categories of household income (1 - less than \$10k to 11 - \$200k or more); and EMPLOY1 (employment status), a nominal variable with eight categories (e.g., employed for wages, retired, unable to work). Missing values (e.g., code 9) were excluded for all disparity variables. The regression model further included key contributing factors to adjust for confounding and assess independent risk. These include _HLTHPL1 (Health Insurance Status), a dichotomous variable ("yes/no") included as a measure of access to care (Anim et al., 2024); EDUCA (Education Level), an ordinal variable with six levels, included as a proxy for socioeconomic status (Walker et al., 2014); _TOTINDA (physical activity during the past 30 days other than job), a dichotomous variable ("yes/no") for physical activity (Chen et al., 2025); and _SMOKER3 (smoking status category), a categorical variable with four levels, included as a key behavioral risk factor (Lu et al., 2024). After applying exclusion criteria for missing data across the outcome, disparity, and all contributing factor variables, the final Analytic Sample size for the full regression model was the total number of respondents with complete, non-missing information, totaling $N \leq 9356$.

Exploratory Data Analysis





Individuals identifying as White have the highest proportion of being diabetes-free. Likewise, those employed for wages exhibit the lowest likelihood of having diabetes compared to other employment status groups. In Kings County, NY, 36% of residents have a household income of under \$50K, the highest proportion among the income categories. These findings indicate that lower socioeconomic status may be associated with an increased risk of diabetes, likely due to limited access to healthcare resources.

An outlier was identified in the variable PA3MIN_(minutes of total physical activity per week), with a recorded value of 19,760 minutes (approximately 329 hours per week). We do not recommend removing this outlier, as it could result from legitimate factors, such as a unit conversion error during data entry. To ensure transparency, we will follow up with the original data source (e.g., the physician or study coordinator) and document this finding in the analysis report.

Table 1 Exploratory Data Analysis: Kings County, NY Demographics

Demographic Characteristic	Total	Source
Total Population (n)	2,561,225	County Health Rankings
Age (%)		
Below 18 years	21.8%	County Health Rankings
18-64 years	62.0%	County Health Rankings
65 years and older	16.2%	County Health Rankings
Sex (%)		
Male	47.6%	County Health Rankings
Female	52.4%	County Health Rankings
Race (%)		
Non-hispanic White	37.2%	County Health Rankings
Non-hispanic Black	28.6%	County Health Rankings
Hispanic	18.9%	County Health Rankings
Asian	13.1%	County Health Rankings
American Indian or Alaska Native	1.0%	County Health Rankings
Native Hawaiian or Other Pacific Islander	0.1%	County Health Rankings
Median Household Income	\$81,027	Census Data
Household Income Brackets/ Income Category		
Under \$50K	36%	Census Data
\$50K - \$100K	24%	Census Data
\$100K - \$200K	24%	Census Data
Over \$200K	16%	Census Data

Education Level		
No Degree	15%	Census Data
High School	22%	Census Data
Some College	19%	Census Data
Bachelor's	25%	Census Data
Post- Grad	18%	Census Data
Health Condition		
Type 2 Diabetes (Prevalence, 2022)	11.6%	CDC PLACES
Coronary Heart Disease (Prevalence, 2022)	6.2%	CDC PLACES
Hypertension (Prevalence, 2021)	32.7%	CDC PLACES
Obesity (prevalence, 2023)	24.3%	NY Health Dept
Priority Health Need		
Food Insecurity (Prevalence, 2023)	30.1 %	NY Health Dept.
Diabetes (primary diagnosis) hospitalizations (per 10,000, age-adjusted, 2020-2022)	19.9	NY Health Dept.

Analytic Plan

This analysis used data from the 2023 Behavioral Risk Factor Surveillance System (BRFSS) for adults residing in the New York Metropolitan Area. The analytic approach included two components: (1) a bivariate health disparity analysis to identify differences in type 2 diabetes (T2DM) prevalence by race/ethnicity, and (2) a multivariable regression analysis to examine factors associated with T2DM after accounting for demographic and behavioral characteristics. The significance level was set at $P < 0.05$ for all analyses.

Bivariate Analysis – Health Disparity Analysis

I compared the prevalence of T2DM (yes/no) across four categories of race/ethnicity: non-Hispanic White, non-Hispanic Black, Hispanic, Mixed, and Other Race. Because both T2DM status and race/ethnicity are categorical variables, I used a chi-square test of independence to test whether T2DM prevalence differed significantly among these groups. The chi-square test was chosen because it determines whether observed group differences in a categorical outcome are statistically significant beyond what would be expected by chance.

Multivariate Analysis – Health Factors

Logistic regression was selected as it is appropriate for modeling a binary outcome and allows multiple covariates to be considered simultaneously. This analytic approach allows estimation of the adjusted odds ratio for T2DM associated with each racial/ethnic group, holding age, sex, education, insurance status and smoking status constant. To explore factors associated with Type 2 Diabetes (T2DM) prevalence, a multivariable logistic regression analysis was conducted using 2023 BRFSS data for the New York Metropolitan Area. The dependent variable (outcome) was

Diabetes status (DIABETE4), coded as yes/no (based on whether participants reported ever being told they had diabetes). The main exposure variables of interest for disparity assessment were General Health (GENHLTH) and Employment Status (EMPLOY1). Covariates included in the final model, selected via a stepwise process, were: self-rated health (GENHLTH), number of days of poor health preventing activities (POORHLTH), access to a personal healthcare provider (PERSDOC3), difficulty paying medical costs (MEDCOST1), high cholesterol status (_RFCHOL3), difficulty walking/climbing (DIFFWALK), difficulty dressing (DIFFDRES), educational attainment (EDUCA), employment status (EMPLOY1), and current e-cigarette use (ECIGNOW2). Logistic regression was the appropriate choice for this analysis, as it effectively models a binary outcome and allows for the simultaneous adjustment of multiple covariates. This analytic approach estimated the adjusted Odds Ratio (OR) for diabetes associated with each exposure variable category, controlling for the independent effects of all other variables included in the model.

Hypotheses

Health Disparity Hypothesis: Individuals from different racial and ethnic groups differ in the prevalence of type 2 diabetes, with some groups experiencing a higher burden than others.

- H_0 : Prevalence of type 2 diabetes (T2DM) does not differ across race/ethnicity groups.
- H_A : Prevalence of type 2 diabetes (T2DM) differs significantly across race/ethnicity groups.

Risk Factor Hypothesis: Individuals from different racial and ethnic groups have higher odds of type 2 diabetes independent of smoking, age, sex, education, and insurance status.

- H_0 : There is no association between self-identified race/ethnicity and the odds of type 2 diabetes after adjustment for smoking, age, sex, education, and insurance status.
- H_A : Self-identified race/ethnicity is independently associated with increased odds of type 2 diabetes after adjustment for smoking, age, sex, education, and insurance status.

Findings

Bivariate Analysis

We examined _RACEGR3 (five-level race/ethnicity) and New York. The statistical test I used was the Chi square test. I previously found there was a significant relationship between race and New York p-value at $p = 2.926e-15$ which is significant because the p-value is less than alpha level of 0.05. This indicates a statistically significant relationship between race and the prevalence of diabetes in the population studied. In Group 1 (White only, Non-Hispanic) , 319 out of 4001 individuals (approximately 7.98%) reported having diabetes, while 3682 (92.02%) reported not having the condition. In contrast, Group 2 (Black only, Non-Hispanic) exhibited the highest prevalence, with 224 out of 1476 individuals (15.15%) reporting diabetes, compared to 1252 (84.85%) without it. Groups 3 (Other race only, Non-Hispanic) and 5 (Hispanic) also showed notable percentages, with 127 out of 1044 (12.16%) and 343 out of 2690 (12.75%) individuals reporting diabetes, respectively. Group 4 (Multiracial, Non-Hispanic) had a lower prevalence of 15 out of 145. According to this information it appears that group 2 (Black only, Non-Hispanic) is the most affected group with diabetes with groups 3 (Other race only, Non-Hispanic) and 5 (Hispanic) following second. The analysis clearly indicates a significant disparity in diabetes prevalence among different racial and ethnic groups in New York. A statistically significant disparity was observed in diabetes prevalence across racial and ethnic

groups in New York ($p < 0.001$), indicating unequal burden. Interpreted through a health equity lens, the significantly higher prevalence among Black, Non-Hispanic (15.15 %), and Hispanic (12.75 %) populations compared to White, Non-Hispanic individuals (7.98 %) reflects structural and social determinants of health, such as systemic barriers to quality care and resources, rather than inherent biological differences.

Group Description	Diabetes	Total Individuals	Prevalence (%)
Group 1: White Only, Non-Hispanic	319	4001	7.98
Group 2: Black Only, Non-Hispanic	224	1476	15.15
Group 3: Other Races Only, Non-Hispanic	127	1044	12.16
Group 4: Multiracial, Non-Hispanic	15	145	10.34
Group 5: Hispanic	343	2690	12.75

Regression Analysis

Individuals with worse health perceptions (Fair, Good, Poor) have significantly higher odds of developing diabetes compared to those with Excellent health. Lack of a personal health care provider is associated with lower odds of diabetes, which might be counterintuitive but can reflect lifestyle or management differences. Poor physical health days have a negative correlation with diabetes, and specific employment statuses (out of work, retired) significantly increase odds of diabetes. Based on the multivariable logistic regression analysis of New York Metro Area data (BRFSS 2023), General Health Perception was the most influential factor associated with self-reported diabetes, exhibiting a highly significant dose-response relationship (as detailed in Table 5). Compared to individuals reporting Excellent health, those reporting poor health had over 31 times higher odds of having diabetes (OR = 31.74; 95% CI: 11.08, 90.96; $p < 0.001$), while those reporting Fair (OR = 10.31; $p < 0.001$) or Good (OR = 6.04; $p < 0.001$) health also faced significantly elevated odds. Other significant positive associations included reporting a history of High Cholesterol (OR = 2.49; $p < 0.001$), being Out of Work for less than a year (OR = 4.11; $p < 0.001$), or being Retired (OR = 3.62; $p = 0.001$), all of which significantly increased the odds of diabetes compared to employed individuals without high cholesterol.

Conversely, two variables showed statistically significant, counter-intuitive associations suggesting potential bias in the data. Individuals reporting No Personal Healthcare Provider (PCP) had significantly lower odds of a diabetes diagnosis (OR = 0.388; 95% CI: 0.185, 0.816; $p = 0.011$), which is likely due to detection bias where individuals with a confirmed chronic disease maintain a PCP for treatment and follow-up. Furthermore, an increase in Poor Health

Days (days restricted by poor health) was associated with a slight decrease in diabetes odds (OR = 0.951 per day; $p = 0.002$). This negative correlation may reflect the better management or reduced severity of general symptoms among those who are already diagnosed and adhering to treatment. These findings highlight the strong role of self-rated health and key demographic factors in diabetes prevalence, while cautioning against simple causal interpretation of access-to-care variables.

Group Description	Total Individuals	Diabetes Prevalence (%)	Disparity (times higher than Group 1)
Group 1: White Only, Non-Hispanic (Reference)	4001	7.98% (319 cases)	1.00
Group 2: Black Only, Non-Hispanic (Most Affected)	1476	15.15% (224 cases)	1.90 times
Group 5: Hispanic	2690	12.75% (343 cases)	1.60 times
Group 3: Other Races Only, Non-Hispanic	1044	12.16% (127 cases)	1.52 times
Group 4: Multiracial, Non-Hispanic	145	10.34% (15 cases)	1.30 times

Discussion

Analysis of Findings

The statistical analysis conducted on the 2023 Behavioral Risk Factor Surveillance System (BRFSS) data for the New York Metropolitan Area yields significant findings regarding the prevalence of Type 2 Diabetes Mellitus (T2DM). The results from both the bivariate and multivariable analyses confirm expected associations with health status and demographics, and, most crucially, highlight the persistence of profound racial and ethnic health disparities. These empirical findings must be interpreted through the lens of a broader epidemiological framework, synthesizing concepts from social determinants of health (SDOH), environmental exposures, and structural inequities, which collectively shape disease burden far more than individual risk factors alone.

The bivariate analysis demonstrated a highly significant association between race/ethnicity and T2DM prevalence ($p < 0.001$), compelling the rejection of the null hypothesis. The concrete prevalence figures reveal a substantial disparity: the prevalence of T2DM among Black, Non-Hispanic individuals was 15.15% (224 out of 1476), nearly double the rate observed in the White, Non-Hispanic population at 7.98% (319 out of 4001). Additionally, Hispanic individuals experienced a high prevalence of 12.75% (343 out of 2690). These unequal burdens are not random variation; they are direct signatures of the impact of Social Determinants of Health

(SDOH). As demonstrated by Walker et al. (2014), factors outside the traditional clinical setting, such as economic stability and neighborhood characteristics, significantly affect critical health markers like glycemic control (HbA1c). In the context of the New York Metropolitan Area, these racial groups are disproportionately affected by poor neighborhood quality, which translates into limited access to quality food (nutritional factors), higher exposure to pollutants (environmental factors), and diminished healthcare access. For instance, the systematic review by Schwingshackl et al. (2017) linked higher consumption of red and processed meats and sugar-sweetened beverages with increased T2DM risk, and communities impacted by poor SDOH often lack affordable access to whole grains and fresh produce, making healthier dietary choices difficult to sustain.

Furthermore, the significant racial disparities must be viewed through the framework of structural racism and its impact on healthcare access and quality. While one might assume these differences disappear after accounting for socioeconomic status, Canedo et al. (2018) showed that disparities in receiving key quality care indicators, such as HbA1c tests, persisted for racial and ethnic minority groups even when controlling for education and insurance. This suggests that systemic issues—like implicit bias in provider interactions and structural barriers—actively contribute to an inequitable standard of care. The systematic review by Anim et al. (2024) directly associated perceived interpersonal discrimination and racism from healthcare professionals with negative T2DM outcomes, specifically for Black Americans, indicating that the quality of interaction mediates self-management and clinical results. Therefore, the higher prevalence rates observed in the Black and Hispanic populations in this analysis reflect the outcome of these structural and interpersonal barriers, rather than inherent biological predispositions.

The multivariable logistic regression analysis provided further context by identifying factors strongly associated with a diabetes diagnosis, independent of other variables in the model. The most powerful association was with General Health Perception, exhibiting a highly significant dose-response relationship. Individuals reporting Poor health had over 31 times higher odds of a diabetes diagnosis (OR=31.74; $p<0.001$) compared to those with Excellent health. This finding underscores the strong correlation between chronic disease and self-reported well-being. Other significant positive associations included having High Cholesterol (OR=2.49), which confirms the clinical link between T2DM and other features of metabolic syndrome. The association with specific employment statuses, such as being Retired (OR=3.62) or Out of Work (OR=4.11), reflects both the typical age demographics of T2DM (middle-aged and older people are at higher risk) and the potential for the disease or its complications (like difficulty walking or kidney disease) to impact an individual's ability to work, thereby highlighting the interplay between disease and economic stability.

Crucially, the multivariable model also revealed counter-intuitive findings related to access-to-care variables that underscore the limitations of using cross-sectional, self-reported data. The result showing that individuals reporting No Personal Healthcare Provider (PCP) had significantly *lower* odds of a diabetes diagnosis (OR=0.388) likely reflects detection bias. A chronic disease like T2DM, often asymptomatic in its preclinical phase, requires a provider for diagnosis. People without a PCP are simply less likely to have been screened and thus less likely to have been *detected* as having diabetes, creating an artificial negative association. This highlights a fundamental challenge in the BRFSS data: it measures diagnosed diabetes, not true prevalence, and cannot establish the temporality necessary for causal inference, as noted by the inability to fulfill Hill's criteria. Similarly, the slight negative correlation with Poor Health Days

may reflect selection or management bias, where individuals with a confirmed diagnosis might be adhering to treatment, thereby reducing the severity of general health-related restrictions. These methodological challenges caution against simple causal interpretations, especially for access variables, and reinforce the need to use longitudinal studies to establish genuine risk factor relationships.

In summary, the statistical findings from the New York BRFSS data are a stark reminder that T2DM prevalence is not an individual failing but a product of overlapping, adverse social and structural conditions. The significant and persistent racial disparities confirm the epidemiological literature on the effects of structural racism and unequal distribution of SDOH. Moving forward, interventions must be multi-pronged, addressing not just clinical management but the deep-seated inequities in nutrition, environment, and healthcare access that are driving the observed burden of disease in the Black and Hispanic communities.

Limitations

One key limitation of the BRFSS dataset is related to its sampling methodology. Since data is collected via telephone interviews, it may exclude individuals without access to phones or those who are less likely to respond to surveys, such as certain low-income populations or younger adults. This can lead to a biased sample that may not fully represent the diversity of the population. Additionally, the reliance on self-reported data introduces potential measurement bias, particularly in responses related to sensitive conditions like diabetes, where individuals may underreport or misinterpret their health status. Furthermore, the cross-sectional nature of the data limits causal inference, as it captures a snapshot in time rather than tracking changes over time or

establishing direct cause-and-effect relationships. Thus cannot establish causality because cross-sectional doesn't fit the temporality criteria of Hill's.

The epidemiological literature identified several critical confounding factors that could impact the interpretation of the BRFSS analysis. In the context of nutrition, while Schwingshackl et al. (2017) adjusted for factors like age, sex, education, and physical activity, they acknowledged that other uncaptured lifestyle factors and dietary habits could still affect outcomes. Similarly, Pollakova et al. (2021) noted that the impact of food quality and preparation methods was not fully assessed, which could influence the outcomes of dietary pattern studies. In the environmental sphere, Yang et al. (2019) controlled for age, sex, and socioeconomic status in their air pollution analysis, but noted that potential unmeasured confounders like genetic predisposition and local healthcare access may still impact results. The confounding impact of socioeconomic status (SES), health insurance status, and behavioral/clinical risk factors was discussed by Anim et al. (2024), and access to healthcare, comorbidities, and behavioral factors were identified as potential confounders by Egede et al. (2022). The influence of cultural factors affecting health-seeking behavior and lifestyle choices was also raised as a neglected confounder in a study on T2DM and dementia.

The potential impact of these unmeasured or partially adjusted confounders on the BRFSS findings is substantial. For instance, the significant racial disparity in T2DM prevalence could be further explained if we could fully adjust for neighborhood-level socioeconomic resources and community-level physical activity resources (like walkability and green space, which are linked to lower risk), data not included in the BRFSS. Without comprehensive data on individual-level dietary quality or long-term exposure to local air pollution (PM_{2.5} and NO₂ are associated with

T2DM), the true independent effect of race/ethnicity, independent of these geographically-linked confounders, remains potentially inflated.

The BRFSS, by nature of being a telephone survey focused on self-reported risk behaviors and chronic conditions, lacks several data points critical for a deep understanding of health disparities related to T2DM. First, the dataset lacks objective clinical and environmental measures since it relies on self-reported diagnosis and not objective clinical markers like measured HbA1c levels, fasting blood glucose, or lipid panels. It also lacks objective, localized environmental data, such as air quality indices or proximity to major roadways , which are known risk factors for T2DM. Second, the BRFSS is missing crucial measures of structural racism and discrimination. While the analysis controlled for race/ethnicity, education, and income, it lacked granular measures of structural racism or perceived interpersonal discrimination within healthcare settings. These factors are key drivers of disparities that persist even after controlling for traditional SES and access variables. Third, in terms of dietary quality and physical activity detail, the BRFSS typically only captures broad dichotomous variables for physical activity (like physical activity in the past 30 days), which lacks the specificity of metrics like walkability or detailed, long-term dietary assessments required to fully explore the nutritional determinants of T2DM risk. Finally, regarding non-clinical SDOH outcomes, while the background literature stresses the importance of cost and quality of life in T2DM management, the BRFSS data is limited in measuring these non-clinical outcomes. The reliance on self-reported, cross-sectional data, coupled with these missing critical data points, limits the ability to isolate the specific mechanisms—be they dietary, environmental, or systemic—through which race/ethnicity exerts its observed influence on T2DM prevalence. Addressing health disparities effectively requires integrating objective

clinical, structural, and environmental data that the current BRFSS cross-sectional design cannot provide.

Conclusion

The statistical analysis yields several fundamental takeaways and insights regarding Type 2 Diabetes Mellitus (T2DM) in the New York Metropolitan Area. Foremost among these is the confirmation of a severe and highly significant health disparity in T2DM prevalence, with the burden resting disproportionately on Black, Non-Hispanic (15.15% prevalence) and Hispanic populations (12.75% prevalence) compared to their White, Non-Hispanic counterparts (7.98% prevalence). This finding is not reflective of inherent biological difference but is a powerful empirical manifestation of structural racism and the unequal distribution of Social Determinants of Health (SDOH) (Canedo et al., 2018; Walker et al., 2014).

A key insight from the multivariable regression is the overwhelming predictive power of self-rated health. The finding that individuals reporting Poor health had over 31 times higher odds of a diagnosis ($OR=31.74$; $p<0.001$) suggests that T2DM is often embedded within a broader, complex pattern of poor overall physical and mental health. This is particularly concerning in communities where poverty is pervasive, such as certain neighborhoods populated by people of color, where access to nutritious food, safe recreational spaces, and primary healthcare resources is often limited (Walker et al., 2014). These systemic barriers not only create challenges for individuals seeking to maintain healthy lifestyles but also contribute to higher rates of T2D risk (Chen et al., 2025). Furthermore, the counter-intuitive findings

regarding healthcare access in the BRFSS data (e.g., lower odds of diagnosis for those without a Personal Healthcare Provider) serve as a crucial methodological insight: analyses based on cross-sectional, self-reported data are heavily prone to detection bias, limiting their capacity to establish causal inference and potentially masking the true prevalence of the undiagnosed disease.

Building upon this knowledge, the immediate next steps for future research and analyses in the New York Metropolitan Area must address these data and methodological gaps. Research should decisively shift from cross-sectional designs to longitudinal studies to establish temporality and better understand the disease's natural history (Nishida & Watada, 2024). Analytically, efforts are needed to integrate more granular data to explain the persistent racial disparities. This involves supplementing self-reported data with objective measures of environmental exposure (e.g., localized air quality indices for PM2.5 and NO2) and structural factors (e.g., neighborhood-level measures of food environment quality and walkability) to better isolate the independent effect of race/ethnicity from geographically-linked confounders (Yang et al., 2019). Furthermore, research must focus on collecting specific data on perceived interpersonal discrimination and provider-level bias within the healthcare system to quantify the mechanism by which systemic inequities translate into negative clinical outcomes for minority patients (Anim et al., 2024).

Based on the strong evidence base identifying modifiable risk factors and systemic barriers, two critical exposures/factors are recommended for policy modification in the New York Metropolitan Area to improve population health in relation to T2DM:

The first recommended exposure to modify is Access to Affordable, High-Quality Nutritional Resources (Dietary Patterns). The epidemiological literature is definitive that high consumption

of red/processed meats and sugar-sweetened beverages increases T2DM risk, while consumption of whole grains and fruits decreases it (Schwingshackl et al., 2017). This correlation emphasizes the critical role that socioeconomic status plays in health outcomes. Modification should involve policy changes such as enhancing access to affordable, healthy food options through partnerships with local grocery stores and farmers' markets to help mitigate the impact of food deserts in low-income areas. Additionally, investing in community wellness programs that promote physical activity—such as free fitness classes and safe green spaces for exercise—can empower residents to adopt healthier behaviors (Chen et al., 2025).

The second recommended factor to modify is the Structural and Interpersonal Quality of Diabetes Care for Minority Groups. The persistence of disparities in receiving key quality care indicators, even after controlling for socioeconomic factors, indicates a failure within the healthcare system itself (Canedo et al., 2018). Modification must directly address systemic issues and provider bias that cause care disparities and negative outcomes for Black and Hispanic patients (Anim et al., 2024). This intervention should include mandatory, sustained anti-bias training for all healthcare professionals in the metropolitan area, coupled with the integration of nonmedical, multicomponent interventions into standard clinical practice, which have been shown to significantly improve HbA1c levels by addressing patients' social needs (Egede et al., 2022). By acknowledging the interconnectedness of socioeconomic factors and health outcomes, and by targeting both the community environment and the quality of clinical care, the metropolitan area can initiate a comprehensive approach to reduce the disproportionate burden of T2DM (Walker et al., 2014).

References

- Alexandre, K., Campbell, J., Bugnon, M., Henry, C., Schaub, C., Serex, M., Elmers, J., Desrichard, O., & Peytremann-Bridevaux, I. (2021). Factors influencing diabetes self-management in adults: An umbrella review of systematic reviews. *JBIC Evidence Synthesis*, 19(5), 1003¹-1118. <https://doi.org/10.11124/JBIES-20-0²0020>
- American Diabetes Association. (n.d.). 2. *Diagnosis and classification of diabetes: Standards of care in diabetes—2025*. Retrieved from https://diabetesjournals.org/care/article/48/Supplement_1/S27/157566/2-Diagnosis-and-Classification-of-Diabetes
- Anim, S. B., Spurlark, R., Turkson-Ocran, R.-A., Bohr, N., Soco, C., & Simonovich, S. D. (2024). A systematic review of the relationship between discrimination, racism, and Type 2 Diabetes healthcare outcomes for Black Americans. *Journal of Racial and Ethnic Health Disparities*, 11(5), 2935–2944. <https://doi.org/10.1007/s40615-023-01751-x>
- Beulens, J. W. J., Pinho, M. G. M., Abreu, T. C., van den Berg, A. P., van Bilsen, J., Gansevoort, R. T., Navis, G., Sijtsma, K., Sluiter, W. J., & Wolffenbuttel, B. H. R. (2022). Environmental risk factors of type 2 diabetes—an exposome approach. *Diabetologia*, 65(2), 263-274. <https://doi.org/10.1007/s00125-021-05618-w>

- Canedo, J. R., Miller, S. T., Schlundt, D., Fadden, M. K., Sanclement, J. C., & Egede, L. E. (2018). Racial and ethnic differences in diabetes quality of care: The role of socioeconomic status. *Journal of General Internal Medicine*, 33(10), 1678–1685.
- Chen, Y., Deng, Y., Wu, M., Ma, P., Pan, W., Chen, W., Zhao, L., & Huang, X. (2025). Impact of pesticides exposure and type 2 diabetes risk: A systematic review and meta-analysis. *Endocrine*, 87(2), 448–458. <https://doi.org/10.1007/s12020-024-04067-w>
- Cho, M.-K., & Kim, M. Y. (2021). Self-management nursing intervention for controlling glucose among diabetes: A systematic review and meta-analysis. *International Journal of Environmental Research and Public Health*, 18(23), 12750. <https://www.mdpi.com/1660-4601/18/23/12750>
- Cooper, I. D., Petagine, L., Soto-Mota, A., Duraj, T., Scarborough, A., Norwitz, N. G., Seyfried, T. N., Furon, M. A., & Kyriakidou, Y. (2025a, September 2). Ketosis suppression and ageing (KetoSAGE): The effect of suppressing ketosis on GKI and liver biomarkers in healthy females. *MDPI*. <https://www.mdpi.com/2673-4389/5/3/41>
- Cooper, I. D., Petagine, L., Soto-Mota, A., Duraj, T., Scarborough, A., Norwitz, N. G., Seyfried, T. N., Furon, M. A., & Kyriakidou, Y. (2025b, September 2). Ketosis suppression and ageing (KetoSAGE): The effect of suppressing ketosis on GKI and liver biomarkers in healthy females. *MDPI*. <https://www.mdpi.com/2673-4389/5/3/41>
- Dendup, T., Feng, X., Clingan, S., & Astell-Burt, T. (2018). Environmental risk factors for developing Type 2 Diabetes Mellitus: A systematic review. *International Journal of*

Environmental Research and Public Health, 15(1), 78.

<https://doi.org/10.3390/ijerph15010078>

Egede, L. E., Walker, R. J., Linde, S., Campbell, J. A., Dawson, A. Z., Williams, J. S., & Ozieh, M. N. (2022). Nonmedical interventions for Type 2 Diabetes: Evidence, actionable strategies, and policy opportunities. *Health Affairs*, 41(7), 963–970.

<https://doi.org/10.1377/hlthaff.2022.00236>

Kahlenberg, H., Jiroutek, M. R., & Misciagno, S. A. (2025). Ethnic and racial disparities in the association between Type II Diabetes Mellitus and dementia. *Journal of Racial and Ethnic Health Disparities*, 12, 41-48. <https://doi.org/10.1007/s40615-023-01848-3>

Lee, W., Lloyd, J. T., Giuriceo, K., Day, T., Shrank, W., & Rajkumar, R. (2020). Systematic review and meta-analysis of patient race/ethnicity, socioeconomics, and quality for adult type 2 diabetes. *Health Services Research*, 55(5), 741-772.

<https://doi.org/10.1111/1475-6773.13326>

Levy, N. K., Park, A., Solis, D., Hu, L., Langford, A. T., Wang, J., et al. (2022). Social determinants of health and diabetic gastroparesis: A retrospective analysis. *JMIR Formative Research*, 6(10), e40164.

Lu, X., Xie, Q., Pan, X., Zhang, R., Zhang, X., Peng, G., Zhang, Y., Shen, S., & Tong, N. (2024). Type 2 diabetes mellitus in adults: Pathogenesis, prevention and therapy. *Signal Transduction and Targeted Therapy*, 9(1), 262.

<https://doi.org/10.1038/s41392-024-01951-9>

- McLaurin, N., Tabibi, D., Wang, T., Alhalimi, T., Lehrer, H. M., Harrison, L., Jr., Tanaka, H., & Steinhardt, M. A. (2024). Coping With Discrimination Among African Americans With Type 2 Diabetes: Factor Structure and Associations With Diabetes Control, Mental Distress, and Psychosocial Resources. *Preventing Chronic Disease*, 21, E06. <https://doi.org/10.5888/pcd21.230189>
- Papamichou, D., Panagiotakos, D. B., & Itsiopoulos, C. (2019). The relationship between Mediterranean Diet and Type 2 Diabetes: The current knowledge. *Nutrition, Metabolism and Cardiovascular Diseases*, 29(6), 531-543. <https://doi.org/10.1016/j.numecd.2019.03.003>
- Parker, E. D., Lin, J., Mahoney, T., Ume, N., Yang, G., Gabbay, R. A., ElSayed, N. A., & Bannuru, R. R. (2024). Economic costs of diabetes in the U.S. in 2022. *Diabetes Care*, 47(1), 26–43. <https://doi.org/10.2337/dci23-0085>
- Pollakova, D., Andreadi, A., Pacifici, F., Della-Morte, D., Lauro, D., & Tubili, C. (2021). The impact of vegan diet in the prevention and treatment of type 2 diabetes: A systematic review. *Nutrients*, 13(6), 2123. <https://doi.org/10.3390/nu13062123>
- Schwingshackl, L., Hoffmann, G., Lampousi, A.-M., Knuppel, S., Iqbal, K., Schwedhelm, C., Bechthold, A., Schlesinger, S., & Boeing, H. (2017). Food groups and risk of type 2 diabetes mellitus: A systematic review and meta-analysis of prospective studies. *European Journal of Epidemiology*, 32(4), 363-375. <https://doi.org/10.1007/s10654-017-0246-y>

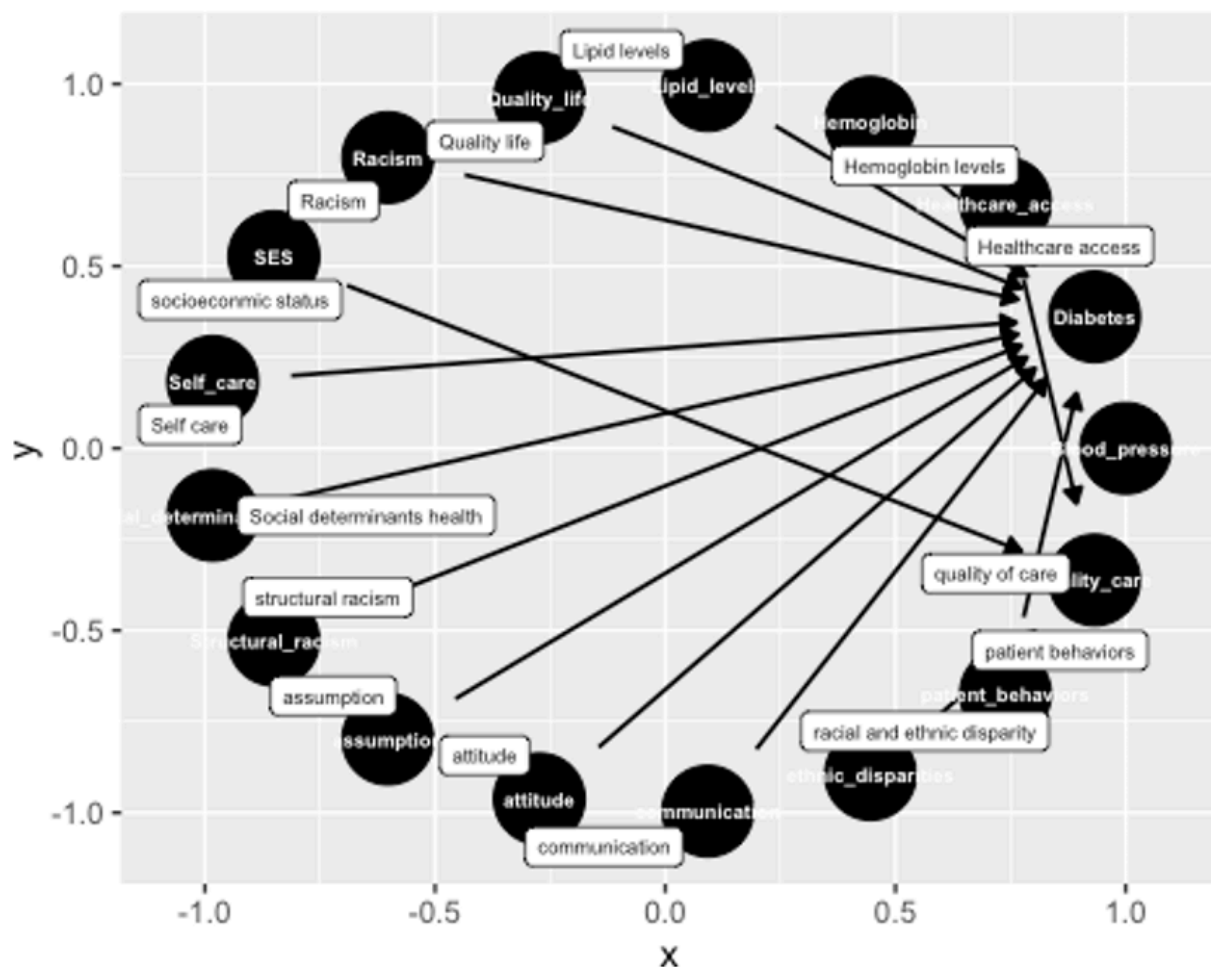
- Titus, S. K., & Kataoka-Yahiro, M. (2019). A systematic review of barriers to access-to-care in Hispanics With Type 2 Diabetes. *Journal of Transcultural Nursing*, 30(3), 280-290.
<https://doi.org/10.1177/1043659618817757>
- U.S. National Library of Medicine. (n.d.). *Type 2 diabetes| adult-onset diabetes*. MedlinePlus.
Retrieved from <https://medlineplus.gov/diabetestype2.html>
- U.S. National Library of Medicine. (n.d.). *Type 2 diabetes: MedlinePlus medical encyclopedia*. MedlinePlus. Retrieved from <https://medlineplus.gov/ency/article/000313.htm>
- Walker, R. J., Smalls, B. L., Campbell, J. A., Strom Williams, J. L., & Egede, L. E. (2014). Impact of social determinants of health on outcomes for Type 2 Diabetes: A systematic review. *Endocrine*, 47(1), 29–48. <https://doi.org/10.1007/s12020-014-0195-0>
- Ward, L. M., Pendergrass, D. B., Chima, C. C., Thorpe, R. J., Bruce, M. A., Blackburn, H. F., Palombo, C. F., & Beech, B. M. (2020). Access to medications among adults with type 2 diabetes using pharmacy- or clinic-based medication assistance programs: A systematic review. *Journal of the American Pharmacists Association*, 60(6), e411–e421.
<https://doi.org/10.1016/j.japh.2020.07.016>
- Yang, M., Cheng, H., Shen, C., Liu, J., Zhang, H., Cao, J., & Ding, R. (2019). Effects of long-term exposure to air pollution on the incidence of type 2 diabetes mellitus: A meta-analysis of cohort studies. *Environmental Science and Pollution Research*, 27, 798-811. <https://doi.org/10.1007/s11356-019-06824-1>
- Zoungas, S., Mohan, V., Amutha, A., Romero-Aroca, P., Gasevic, D., Magliano, D. J., Teede, H. J., Chalmers, J., & Zoungas, S. (2021). Impact of age at type 2 diabetes mellitus

diagnosis on mortality and vascular complications: systematic review and meta-analyses.

Diabetologia, 64, 275–287. <https://doi.org/10.1007/s00125-020-05319-w>

Appendices

Appendix 1. Causal Diagram



Appendix 2. County Profile

Table 1: Kings County, NY Demographics

Demographic Characteristic	Total	Source
Total Population (n)	2,561,225	County Health Rankings
Age (%)		
Below 18 years	21.8%	County Health Rankings
18-64 years	62.0%	County Health Rankings
65 years and older	16.2%	County Health Rankings
Sex (%)		
Male	47.6%	County Health Rankings
Female	52.4%	County Health Rankings
Race (%)		
Non-hispanic White	37.2%	County Health Rankings
Non-hispanic Black	28.6%	County Health Rankings
Hispanic	18.9%	County Health Rankings
Asian	13.1%	County Health Rankings
American Indian or Alaska Native	1.0%	County Health Rankings

Native Hawaiian or Other Pacific Islander	0.1%	County Health Rankings
Median Household Income	\$81,027	Census Data
Health Condition		
Type 2 Diabetes (Prevalence, 2022)	11.6%	CDC PLACES
Coronary Heart Disease (Prevalence, 2022)	6.2%	CDC PLACES
Hypertension (Prevalence, 2021)	32.7%	CDC PLACES
Obesity (prevalence, 2023)	24.3%	NY Health Dept
Priority Health Need		
Food Insecurity (Prevalence, 2023)	30.1 %	NY Health Dept.
Diabetes (primary diagnosis) hospitalizations (per 10,000, age-adjusted, 2020-2022)	19.9	NY Health Dept.

Table 2: Kings County, NY Determinants of Health

Factor	Value for Kings County	Source
Depressive Symptoms in Adults (Prevalence, 2022)	20.7%	CDC PLACES
Feeling Socially Isolated Among Adults (Prevalence, 2022)	31.9%	CDC PLACES

Lack of Health Insurance in Adults 18-64 (Prevalence, 2022)	10.8%	CDC PLACES
Children in Poverty (Proportion, 2019-2023)	27%	County Health Rankings
Air pollution: particulate Matter	8.4	County Health Rankings
Average daily density of fine particulate matter in micrograms per cubic meter (PM2.5)	8.4	County Health Rankings
No Leisure-Time Physical Activity in Adults (Prevalence, 2022)	23.7%	CDC PLACES
Rental Households Above Affordability (Proportion, 2013-2017)	52.2%	NY Comptroller Office
Rental Households Above Severe Cost Burden Thresholds (Proportion, 2013-2017)	29.2%	NY Comptroller Office
Owner Households Above Affordability (Proportion, 2013-2017)	40.4%	NY Comptroller Office
Owner Households Above Severe Cost Burden Thresholds (Proportion, 2013-2017)	21.8%	NY Comptroller Office

Table 3: Community Resources & Assets - Kings County, NY

Resource / Asset	Count / Data	Source
Tree Canopy Cover (Proportion, 2023)	18%	Trees equity

Traffic Volume (Average traffic volume per meter of major roadways in the county, 2020)	633	County Health Rankings
Primary care physician:providers to patient (Ratio, 2025)	1,470:1	County Health Rankings
Mental health providers:patient (Ratio, 2025)	310:1	County Health Rankings
Preventable hospital stays:patients (Count, 2025)	2,975	County Health Rankings

Table 4: Climate Impacts on Region - Kings County, NY

Climate Event or Anticipated Climate Event	Available Data (Current State)	Current/Recent Impact (optional for Fall)	Anticipated/Future Impact (optional for Fall)	Source
Drought	Approaching drought status for last 3 years	Challenges sourcing water for crops	If ongoing, increased risk to food systems	NIDIS Drought
Flooding	Increased heavy rainfall events	Disruptions to food supply chains and access to clinics, making diabetes management harder	Continued flooding could increase food insecurity and hinder access to care for people with diabetes	USDA, 2022, IPCC, 2023, NYC EH Data portal Flooding

Heat	Rising frequency of extreme heat events in last 5 years	Increased heat stress can exacerbate complications in individuals with type 2 diabetes	More frequent and intense heatwaves could increase hospitalization risk and exacerbate diabetes-related complications	CDC, 2023; WHO, 2022, NYC EH Data portal Heat
Winter Storms	Escalating frequency and intensity of nor'easter / winter storms	Transportation disruptions, power outages (affecting medication storage), heightened cold exposure risks for vulnerable groups	Variable winter may continue to produce severe storms, overwhelming emergency services and endangering those with chronic illnesses	NPCC, 2024 NOAA, 2023 Winter
Poor Air Quality Days	Poor air quality days have been increasing, particularly during heat events and near high-traffic areas	Worsened cardiovascular complications in individuals with type 2 diabetes, increased inflammation from air pollutants	More frequent poor air quality episodes could increase diabetes-related hospitalizations and complications	NPCC, 2024 EPA, 2023 NYC DOHMH, 2022 Poor air quality

Appendix 3. Data Collection and Methods Planning

Feel social isolation	How do residents feel about their social connections and support systems?	Qualitative	Structured interview
Child in poverty	How do families in poverty perceive their access to resources and support?	Qualitative	Structured interviews with affected families
Air pollution	What are residents' perceptions of air quality and its health impacts?	Qualitative	Community surveys regarding perceptions and concerns
Average density of fine matter	How does awareness of fine particulate pollution affect health behaviors?	Quantitative	Surveys measuring awareness and behavior
Rental household above affordability	What percentage of rental households struggle to afford housing costs?	Quantitative	Analysis of census data combined with survey data
Above severe cost burden	How do residents feel about their financial strain related to housing?	Qualitative	Focus groups discussing financial burdens and housing
Owner household above affordability	Are homeowners aware of the affordability issues in their area?	Qualitative	Community interviews with homeowners

Severe cost burden thresholds	How does housing cost affect residents' health conditions?	Quantitative	Surveys examining cost burden vs. health outcomes
Tree canopy cover	How important do residents think green spaces are for their health?	Qualitative	Focus groups focused on environmental health
Traffic volume	How do residents perceive traffic impact on their daily lives?	Qualitative	Community interviews regarding traffic and health
Preventable hospital stays	What factors do residents believe contribute to avoidable hospital visits?	Qualitative	Structured interviews with patients & health providers
Drought	What are residents' experiences and concerns regarding drought conditions?	Qualitative	Community surveys focusing on drought impact
Flooding	How do community members prepare for potential flooding?	Qualitative	Focus groups discussing preparedness and awareness
Heat	How do residents adapt to extreme heat, and what resources do they need?	Qualitative	In-depth interviews exploring coping mechanisms
Winter storms	How do residents plan for winter storms, and what resources do they consider essential?	Qualitative	Focus groups on winter preparedness

Poor air quality days	How does poor air quality affect residents' daily activities and health?	Qualitative	Community surveys assessing impact of air quality
----------------------------------	--	-------------	---